Finite terms: check, DGLAP, running coupling, and the form of AABKL

1 Answer

Your file `Finite_terms.tex` should be every row of `Finite_terms_1.tex` minus its JIMWLK ($\log \frac{\vee}{\Lambda}$, `Evolution_terms.tex`) and BFKL ($\log \frac{\vee}{k^+}$, `bfkl.tex`) terms.

- **20 of 27 equations are correct as written** (Sec. 2). Each is identical, term by term, to the remaining terms I obtained by exact substitution (`Finite_terms_split.pdf`).
- **Changes** (Sec. 3; your file in red/blue: `Finite_terms_corrected.pdf`):
  - C1 Group II Row IV: the line $-\log \frac{\vee}{k^+} H_4$ is missing. Without it, $\log \frac{\vee}{k^+} H_4$ is counted twice: once in the BFKL list and once inside the ξ integral of line 1.
  - C2 Group III Rows III (both parts), V, VI: $\left[\frac{1}{p^+}\right]_+ \to \left[\frac{1}{p^+}\right]_+^{(k^+)}$ (still unevaluated). Otherwise $-F(0)\log \frac{\vee}{k^+}$ is counted twice.
  - C3 Group III Rows III (both parts), V, VI: subtract the terms (B) of `BFKL_final.pdf` (region $y' \to z$), which are now in the BFKL list. One blue line per row.
  - C4 Group I Row III: remove the two edge-kernel terms (first term of d^3N_{3a}, last term of d^3N_{3b}). The edge kernel is replaced by $\mathcal{D}$ (fix (A)). The last term of d^3N_{3b} is not integrable at $z \to w$. In its place belongs the finite part of the $\mathcal{D}$ region, which is not yet computed (Q3).
  - C5 Group III Row III part 2: the measure is $\int_{w,x',y',z,x}$.
- **Running coupling and DGLAP** are in three rows: Group I Row II part 1, Group I Row V 2b UV, Group II Row IV. Re-checked for your file (`checks_rc.json`, all zero):
  - The β_0 terms of Row II part 1 ($\frac{11}{3}$) and Row V 2b UV ($-\frac{11}{6}$) cancel.
  - The $\frac{\beta_0}{2}\delta(1-\xi)$ of P_{gg} comes from the coupling counterterm of LO.
  - What is left of the running coupling is $\frac{\alpha_s\beta_0}{4\pi}\log\frac{\mu_R^2}{\mu_F^2}\mathrm{LO} + \frac{\alpha_s}{2\pi}N_c\big(\frac{67}{9}-\frac{\pi^2}{3}\big)\mathrm{LO}$ (Sec. 4).
  - DGLAP is $-\frac{\alpha_s}{2\pi}\int_{k^+/\vee}^1 \frac{d\xi}{\xi^2} P^{(r)}_{gg,+}\mathrm{LO}_\xi[\ell_\Delta(\mu_F)] - \log\frac{\vee}{k^+}H_4$ (Sec. 5).
- **Finite terms** (Sec. 6): your rows with C1–C5, without the running coupling and the DGLAP term. They are finite for $\epsilon \to 0$, $\vee \to \infty$, $\Lambda \to 0$, apart from the open finite part of $\mathcal{D}$ (Q3). The result is Sec. 10.
- **Resemblance with AABKL** (Sec. 11). With $\xi = \frac{p^+}{p^+ + k^+}$, the momentum fraction of the unobserved gluon:
  - Group III becomes their real term, $\int_0^{1-x_p} d\xi$ with $x_p \leftrightarrow \frac{k^+}{\vee}$. In the conjugate amplitude the parent crosses the target at $y' = (1-\xi)u + \xi z$, which is their $(1-\xi)\bar{y} + \xi z$.
  - Your $\left[\frac{1}{p^+}\right]_+^{(k^+)}$ together with your $\log(1-\frac{k^+}{\vee})$ lines is exactly their $\frac{1}{[\xi]_+}$.
  - The subtraction C3 is a plus prescription at $\xi \to 1$; what it takes out is the BFKL term.
  - Group I Row V is their virtual term, $\int_0^1 d\xi$.
  - Group II Row IV is their after-target collinear term (DGLAP).
  - JIMWLK is their last ($\int \frac{d\xi}{\xi}$) term.

  Sec. 11 writes your Group III and Group I Row V in these variables, and the result in the layout of their Eq. (4.22).
- $\log\frac{k^+}{\Lambda}$ **and** $\log\frac{\vee}{k^+}$**.** Your rows give $\log\frac{\vee}{\Lambda}\mathcal{J} + \log\frac{\vee}{k^+}(\mathcal{B}-\mathcal{J}) = \log\frac{k^+}{\Lambda}\mathcal{J} + \log\frac{\vee}{k^+}\mathcal{B}$ (`BFKL_final.pdf`). JIMWLK multiplies $\log\frac{k^+}{\Lambda}$ and BFKL multiplies $\log\frac{\vee}{k^+}$.
  - The extra gluon with $\Lambda < p^+ < k^+$ is softer than the measured gluon. It evolves the dense target (JIMWLK).
  - With $k^+ < p^+ < \vee$ the measured gluon is the softer one. The extra gluon then evolves the dilute projectile (linear, BFKL).

  The finite terms do not depend on how the logarithms are grouped.

Notation (as before). $a \equiv \frac{k^+}{\vee}$, $\beta_0 = \frac{11}{3}N_c$ ($n_f = 0$), $d_\perp = 2-\epsilon$, $c_0 \equiv 2e^{-\gamma_E}$, $\ell_r \equiv \log\frac{r^2\mu^2}{c_0^2} = 2\gamma_E + \log\frac{r^2\mu^2}{4}$, $\hat{E}_r \equiv \frac{2}{\epsilon} + \ell_r$; $X = x - y$, $X' = x' - y'$, $\Delta = y' - y$, $\mathcal{K}(r) = \frac{r}{r^2}$, $\mathcal{N} = \frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5}\frac{1}{k^+}$. LO in your notation, and LO with a weight f:

$$\mathrm{LO}[f] \equiv \frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{1}{k^+}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\, f\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\left[U^{e'c}(y') - U^{e'c}(x')\right]\left[U^{ec}(y) - U^{ce}(x)\right]\rho^{e'}(x')\rho^{e}(x),$$

$\mathrm{LO} \equiv \mathrm{LO}[1]$; $\mathrm{LO}_\xi[f]$ is the same with $e^{-ik\cdot(y'-y)} \to e^{-ik\cdot(y'-y)\frac{1}{\xi}}$. $P^{(r)}_{gg,+}(\xi) = 2N_c\big[\frac{\xi}{(1-\xi)_+} + \frac{1-\xi}{\xi} + \xi(1-\xi)\big]$ and $P_{gg} = P^{(r)}_{gg,+} + \frac{\beta_0}{2}\delta(1-\xi)$.

2 Check of your file

Each `align` of your file is compared, term by term, with the same equation of the remaining terms in `Finite_terms_split.pdf`, Secs. 7–9 (`compare_file.py`, after removing spacing and `\notag`). Those remaining terms were obtained from `Finite_terms_1.tex` by exact substitutions, each checked numerically (`../bfkl_split/split_logs.json`).

equation	result	change
Group I Row II(1)	identical	—
Group I Row II(2b)	identical	—
Group I Row II R^i	identical	—
Group I Row III N1	identical	—
Group I Row III N2	identical	—
Group I Row III N3a	identical	C4 (first term)
Group I Row III N3b	identical	C4 (last term)
Group I Row III A	identical	—
Group I Row IV	identical	—
Group I Row V(1)	identical	—
Group I Row V(2a)	identical	—
Group I Row V(2b UV)	identical	—
Group I Row V(2b fin)	identical	—
Group II Row I(1)	identical	—
Group II Row I(2a)	identical	—
Group II Row I(2b)	identical	—
Group II Row I(2c)	identical	—
Group II Row I(2d)	identical	—
Group II Row I(2e)	identical	—
Group II Row I(2f)	identical	—
Group II Row IV	differs: the $-\log\frac{\vee}{k^+}H_4$ line is missing	C1
Group III Row I	identical	—
Group III Row II	identical	—
Group III Row III(1)	differs: $^{(k^+)}$ missing	C2, C3
Group III Row III(2)	differs: $^{(k^+)}$ missing	C2, C3, C5
Group III Row V	differs: $^{(k^+)}$ missing	C2, C3
Group III Row VI	differs: $^{(k^+)}$ missing	C2, C3

The comparison checks C1 and C2 (they are differences from the remaining terms). C3 and C4 come from `BFKL_final.pdf`: the remaining terms of `Finite_terms_split.pdf` were built before (A) and (B) were found, so these rows are "identical" but still need the change. C5 is a typo that both files carry from `Finite_terms_1.tex`.

3 The changes

C1. Group II Row IV: the line $-\log\frac{\vee}{k^+}H_4$

At $\xi \to \frac{k^+}{\vee}$ the phase $e^{-ik\cdot(y'-y)\frac{1}{\xi}}$ of line 1 probes $|y'-y| \sim \frac{k^+}{\vee|\mathbf{k}|}$, and the large-momentum tail of $\log(y'-y)^2$ gives $\log\frac{\vee}{k^+}H_4$, with

$$\log\frac{\vee}{k^+}H_4 = \frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^5}\frac{1}{k^+}\log\frac{\vee}{k^+}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\,\frac{(w-z)\cdot(w'-z)}{(w-z)^2\,(w'-z)^2}\,\frac{(x-z)\cdot(x'-z)}{(x-z)^2\,(x'-z)^2}$$
$$\times\left[U^{e'c}(z) - U^{e'c}(x')\right]\left[U^{ec}(z) - U^{ce}(x)\right]\rho^{e'}(x')\rho^{e}(x). \tag{3.1}$$

Here the measured gluon and its conjugate ($y, y' \to z$) are emitted by the soft gluon z. This term is one of the BFKL terms (`Finite_terms_split.pdf`, Sec. 3; `BFKL_final.pdf`). So the finite row must carry $-\log\frac{\vee}{k^+}H_4$, which is what `Finite_terms_split.pdf` has. Without it the row grows like $\log\frac{\vee}{k^+}$. Together with lines 1 and 2 it is part of the DGLAP term, Eq. (5.1), which is then finite for $\vee \to \infty$.

C2. Group III: $\left[\frac{1}{p^+}\right]_+ \to \left[\frac{1}{p^+}\right]_+^{(k^+)}$

$$\int_0^{\vee-k^+} dp^+ \left[\frac{1}{p^+}\right]_+^{(k^+)} F(p^+) \equiv \int_0^{\vee-k^+} dp^+ \frac{F(p^+) - F(0)\frac{k^+}{p^++k^+}}{p^+} = \int_0^{\vee-k^+} dp^+ \left[\frac{1}{p^+}\right]_+ F(p^+) + F(0)\log\frac{\vee}{k^+}. \tag{3.2}$$

$-F(0)\log\frac{\vee}{k^+}$ is the BFKL term of the row. Your BFKL list (`bfkl.tex`) has it, so with $\left[\frac{1}{p^+}\right]_+$ it is counted twice. With $\left[\frac{1}{p^+}\right]_+^{(k^+)}$ the integral is finite for $\vee \to \infty$ (`../bfkl_split/pplus_check.json`). The term is still not evaluated. In the variables of AABKL, $\left[\frac{1}{p^+}\right]_+^{(k^+)}$ is the plus prescription $\frac{1}{[\xi]_+}$ (Sec. 11).

C3. Group III: the terms (B)

After $\int_{y'}$ (Fourier lemma, `BFKL_final.pdf`),

$$\int_{y'} e^{-ik\cdot(y'-w)}\, e^{-ik\cdot(y'-z)\frac{p^+}{k^+}} \frac{(y'-z)^m}{(y'-z)^2}\, G_m(y') \xrightarrow{p^+\gg k^+} \frac{k^+}{p^++k^+} \int_{y'} e^{-ik\cdot(y'-w)} \frac{(y'-z)^m}{(y'-z)^2}\, G_m(z) + O\Big(\frac{1}{p^{+2}}\Big).$$

In the p^+ bracket, the terms that tend to a constant are $-\frac{1}{k^+}\delta_{im}\delta_{jk'}$ (Rows III(1), V, VI) and $-\frac{p^+}{k^{+2}}[\dots]\frac{1}{p^+(x-z)^2+k^+(x-w)^2} \to -\frac{1}{k^{+2}(x-z)^2}[\dots]$ (Row III(2)). They give $\int_0^{\vee-k^+}\frac{dp^+}{p^++k^+} = \log\frac{\vee}{k^+}$, and that is the term (B) of the row. With the terms (B) in the BFKL list, each row gets one blue line, e.g. Row V:

$$+\ \frac{1}{(2\pi)^3}\frac{ig^4\, f^{c'ca}}{4\pi^4}\frac{1}{k^+}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\frac{1}{p^++k^+}\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\frac{(y'-z)\cdot(y-w)\,(x'-z)\cdot(x-z)}{(y'-z)^2\,(y-w)^2\,(x'-z)^2\,(x-z)^2}$$
$$\times\Big[U^{e'c'}(z)\,\rho^{e'}(x') - U^{c'e'}(x')\,\rho^{e'}(x')\Big]\Big[U^{ce}(x)\,U^{ad}(y)\,\rho^e(x)\rho^d(y) - U^{ad}(w)\,U^{ce}(x)\,\rho^e(x)\rho^d(y)\Big]. \tag{3.3}$$

Its p^+ integral is $-$(term (B)). At large p^+ it cancels the $-\frac{1}{k^+}\delta_{im}\delta_{jk'}$ term after $\int_{y'}$. Each alone grows like $\log\frac{\vee}{k^+}$; the sum is finite (`checks_xi.json`: for a model integrand, $-5.13, -9.75, -14.35$ alone vs. $-0.7896, -0.7999, -0.8000$ together, at $\vee/k^+ \approx 10^2, 10^4, 10^6$). In the variables of AABKL the sum is a plus prescription at $\xi \to 1$ (Sec. 11). The lines of Rows III(1), III(2), VI are in `Finite_terms_corrected.pdf` and in Sec. 6.

C4. Group I Row III: the edge kernel

The first term of d^3N_{3a} and the last term of d^3N_{3b} carry the edge kernel $\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2(z-w)^2}$. The $\log\frac{\vee}{\Lambda}$ and $\log\frac{\vee}{k^+}$ terms of this kernel are replaced by $\mathcal{D}$:

$$\mathcal{D} = -2i\mathcal{N}\int e^{-ik\cdot(w'-w)}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)$$
$$\times f^{abd}\big[U^{ab'}(x') - U^{ab'}(w')\big]\rho^{b'}(x')\,U^{de}(x)\rho^e(x)\,\big[U^{bc}(w) - U^{bc}(y)\big]\rho^c(y) \ +\ \text{c.c.}$$

This is fix F3 (in `Evolution_terms.tex`) and fix (A) (`BFKL_final.pdf`). Without it the three-ρ terms do not cancel. The finite terms of the edge kernel therefore do not belong to the row either.

- The first term of d^3N_{3a} is $\propto \log(1-\frac{\Lambda}{k^+}) \to 0$.
- The last term of d^3N_{3b} is, for $\Lambda \to 0$, $-\frac{ig^4 f^{abd}}{16\pi^5}\frac{1}{k^+}\log\frac{(z-w)^2}{(x-w)^2}\times$ (edge kernel). Near $z \to w$ it behaves like $\frac{\log(1/|z-w|)}{(z-w)^2}$, so $\int_z$ diverges.
- Numerically, with R = Row III of your file at $\vee = 10^{10}$, $\Lambda \to 0$, minus its $\log\frac{\vee}{k^+}$ terms (`edge_check.json`, angular average over z on a circle of radius r around w):

r	$r^2\langle R\rangle$	$r^2\langle R\rangle/\log\frac{1}{r}$	without the last term of d^3N_{3b}: $r\langle R\rangle$
10^{-1}	6×10^{-4}	2×10^{-4}	4×10^{-5}
10^{-2}	1×10^{-3}	2×10^{-4}	9×10^{-6}
10^{-3}	1×10^{-3}	2×10^{-4}	1×10^{-6}
10^{-4}	2×10^{-3}	2×10^{-4}	2×10^{-7}

  $r^2\langle R\rangle/\log\frac{1}{r}$ tends to a constant, so the row is not integrable. Without the term, $r\langle R\rangle \to 0$ and it is integrable.
- On the circle $(x-z)^2 = (x-w)^2$ the $\frac{1}{[(x-z)^2-(x-w)^2]^n}$ of the other terms cancel among themselves; $|R|$ stays finite (`edge_check.json`).
- **Open (Q3):** the finite part of the $\mathcal{D}$ region replaces the red terms. It needs the p^+ integral of Row III in the region where the gluon of $\mathbb{A}^{(1)}(p^+,z)$ becomes the measured gluon, done with the kernel of $\mathcal{D}$. That integral is not in your notes or in mine.

C5. Group III Row III part 2: the measure

The row has $\rho(x')\rho(x)$, and its kernels depend on w, x', y', z, x. So $\int_{w,w',y',z,y,x} \to \int_{w,x',y',z,x}$ (both lines).

4 Running coupling

The β_0 terms. Two rows have β_0: the $\frac{11}{3}$ of Group I Row II part 1 and the $-\frac{11}{6}$ of Group I Row V 2b UV:

$$\frac{d^3N_1^{\beta_0}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4\,N_c}{32\pi^4}\frac{1}{k^+}\frac{11}{3}\int_{w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x'-w')\cdot(x-w)}{(x'-w')^2\,(x-w)^2}\left[-\frac{2}{\epsilon}-2\gamma-\log\left[\frac{(x-w)^2\,\mu^2_{\overline{MS}}}{4}\right]\right]$$
$$\times\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\big[U^{ab}(w)\,\rho^b(x)-U^{ab}(x)\,\rho^b(x)\big]\quad+\text{c.c.}\;=\;\frac{\alpha_s\beta_0}{4\pi}\,\mathrm{LO}\big[\hat{E}_X+\hat{E}_{X'}\big], \tag{4.1}$$

$$\frac{d^3N_{2b\,\mathrm{UV}}^{\beta_0}}{d^3k} = -\frac{1}{(2\pi)^3}\frac{g^4N_c}{16\pi^4}\frac{1}{k^+}\frac{11}{6}\int_{y,z,x',w'} e^{-ik\cdot(w'-z)}\left[\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y-z)^2\mu^2_{\overline{MS}}}{4}\right]\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}$$
$$\times\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[U^{af}(y)-U^{af}(z)\Big]\rho^{b'}(x')\,\rho^f(y)\quad+\text{c.c.}\;=\;-\frac{\alpha_s\beta_0}{4\pi}\,\mathrm{LO}\big[\hat{E}_X+\hat{E}_{X'}\big]. \tag{4.2}$$

With $y\to x$, $z\to w$ in (4.2), the measure, the phase, the kernel and the bracket are those of (4.1), the colour factor changes sign, and $\frac{g^4N_c}{16\pi^4}\frac{11}{6}=\frac{g^4N_c}{32\pi^4}\frac{11}{3}$. So

$$\boxed{\frac{d^3N_1^{\beta_0}}{d^3k}+\frac{d^3N_{2b\,\mathrm{UV}}^{\beta_0}}{d^3k}=0} \tag{4.3}$$

for every a: pole, γ_E, μ and $\log(x-w)^2$ all cancel (`checks_rc.json`). The virtual correction (Row II) and the real emission at short distance (Row V 2b UV) carry opposite β_0 terms, and no $\beta_0\log(x-w)^2$, i.e. no running of the coupling at the emission distance, is left.

The coupling of LO. LO is $\propto g^2$. In $\overline{MS}$, $g^2\to g^2(\mu_R)\big[1-\frac{\alpha_s\beta_0}{4\pi}\frac{2}{\epsilon}\big]$ ($d_\perp=2-\epsilon$), which adds

$$\left.\frac{d^3N}{d^3k}\right|_{\mathrm{CT}} = -\frac{\alpha_s\beta_0}{4\pi}\frac{2}{\epsilon}\,\mathrm{LO} = \frac{\alpha_s}{2\pi}\Big(-\frac{2}{\epsilon}\Big)\frac{\beta_0}{2}\,\mathrm{LO}. \tag{4.4}$$

This pole is the $\frac{\beta_0}{2}\delta(1-\xi)$ part of the DGLAP pole (Sec. 5). After the coupling is renormalized at μ_R and the fragmentation function at μ_F (both $\overline{MS}$), the only β_0 term of the result is the renormalization-group term

$$\boxed{\left.\frac{d^3N}{d^3k}\right|_{\beta_0} = \frac{\alpha_s\beta_0}{4\pi}\,\log\frac{\mu_R^2}{\mu_F^2}\,\mathrm{LO}\Big|_{\alpha_s(\mu_R)}} \tag{4.5}$$

It makes the result independent of μ_R: $\frac{d}{d\log\mu_R^2}\mathrm{LO}|_{\alpha_s(\mu_R)}=-\frac{\alpha_s\beta_0}{4\pi}\mathrm{LO}$. It vanishes for $\mu_R=\mu_F$; then LO is taken with $\alpha_s(\mu_F)$.

The constant. The non-logarithmic part of the bracket of Row II part 1, $-\frac{67}{9}+\frac{\pi^2}{3}$, gives

$$\frac{1}{(2\pi)^3}\frac{g^4N_c}{32\pi^4}\frac{1}{k^+}\int_{w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x'-w')\cdot(x-w)}{(x'-w')^2\,(x-w)^2}\Big[-\frac{67}{9}+\frac{\pi^2}{3}\Big]$$
$$\times\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\big[U^{ab}(w)\,\rho^b(x)-U^{ab}(x)\,\rho^b(x)\big]+\text{c.c.}\;=\;\boxed{\frac{\alpha_s}{2\pi}\,N_c\Big[\frac{67}{9}-\frac{\pi^2}{3}\Big]\mathrm{LO}} \tag{4.6}$$

$N_c(\frac{67}{9}-\frac{\pi^2}{3})=2K$ with $K=N_c(\frac{67}{18}-\frac{\pi^2}{6})$, the constant of the two-loop cusp anomalous dimension that the CMW scheme absorbs into the coupling. It is the finite companion of β_0 in this row.

5 DGLAP

All poles. The poles of your rows, in units of LO (`checks_rc.json`):

G1 Row II part 1 + c.c.	$\frac{\alpha_s N_c}{4\pi}\frac{4}{\epsilon}\big[\frac{11}{3}+\log(1-a)\big]\mathrm{LO}$
G1 Row V 2b UV + c.c.	$\frac{\alpha_s N_c}{2\pi}\frac{4}{\epsilon}\big[-\frac{11}{6}+2\log(1-\frac{\Lambda}{k^+})\big]\mathrm{LO}\quad(\log(1-\frac{\Lambda}{k^+})\to 0)$
G2 Row IV, lines 1, 2	$\frac{\alpha_s}{2\pi}\big(-\frac{2}{\epsilon}\big)\int_a^1\frac{d\xi}{\xi^2}P^{(r)}_{gg,+}(\xi)\,\mathrm{LO}_\xi$
coupling counterterm (4.4)	$\frac{\alpha_s}{2\pi}\big(-\frac{2}{\epsilon}\big)\frac{\beta_0}{2}\mathrm{LO}$
sum	$\frac{\alpha_s}{2\pi}\big(-\frac{2}{\epsilon}\big)\Big[\int_a^1\frac{d\xi}{\xi^2}P_{gg}(\xi)\,\mathrm{LO}_\xi-N_c\log(1-a)\,\mathrm{LO}\Big]$

The first term is the DGLAP pole: it is absorbed in $D_{h/g}(\zeta,\mu_F)$ ($\overline{MS}$), with the convolution $\int_{k^+/\vee}^1\frac{d\zeta}{\zeta^2}D_{h/g}(\zeta,\mu_F)\frac{d^3N_g}{d^3k}(\frac{k^+}{\zeta},\frac{\mathbf{k}}{\zeta})$. The second is not (end of this section).

The DGLAP term. Group II Row IV, lines 1 and 2 of your file, with the pole removed (your bracket $\big[-\frac{d_\perp}{\epsilon}-d_\perp\gamma_E-\frac{d_\perp}{2}\log\frac{(y'-y)^2\mu^2}{4}-1\big]=-\hat{E}_\Delta+O(\epsilon)$ becomes $-\ell_\Delta$; the -1 cancels against $-\frac{d_\perp}{\epsilon}$), together with the log $\frac{\vee}{k^+}H_4$ subtraction (C1):

$$\begin{aligned}\frac{d^3N_{\rm DGLAP}}{d^3k}&=\frac{g^2}{4\pi^2}\,\frac{N_c}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{1}{k^+}\int_{x,y,x',y'}\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left[-2\gamma_E-\log\left(\frac{(y'-y)^2\,\mu_F^2}{4}\right)\right]\\
&\times\left[\int_{\frac{k^+}{\vee}}^1\frac{d\xi}{\xi^2}\,e^{-ik\cdot(y'-y)\frac{1}{\xi}}\left[\xi(1-\xi)+\frac{1-\xi}{\xi}\right]+\int_{\frac{k^+}{\vee}}^1 d\xi\,\frac{\frac{1}{\xi}e^{-ik\cdot(y'-y)\frac{1}{\xi}}-e^{-ik\cdot(y'-y)}}{1-\xi}\right.\\
&\left.+\log\left(1-\frac{k^+}{\vee}\right)e^{-ik\cdot(y'-y)}\right]\\
&\times\Big[U^{e'c}(y')-U^{e'c}(x')\Big]\,[U^{ec}(y)-U^{ce}(x)]\,\rho^{e'}(x')\,\rho^{e}(x)\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4N_c}{8\pi^5}\,\frac{1}{k^+}\,\log\Big[\frac{\vee}{k^+}\Big]\int_{x,x',z,w,w'}e^{-ik\cdot(w'-w)}\,\frac{(w-z)\cdot(w'-z)}{(w-z)^2(w'-z)^2}\,\frac{(x-z)\cdot(x'-z)}{(x-z)^2(x'-z)^2}\\
&\times\Big[U^{e'c}(z)-U^{e'c}(x')\Big]\,[U^{ec}(z)-U^{ce}(x)]\,\rho^{e'}(x')\,\rho^{e}(x)\\
&=\boxed{-\frac{\alpha_s}{2\pi}\int_{k^+/\vee}^1\frac{d\xi}{\xi^2}\,P^{(r)}_{gg,+}(\xi)\,{\rm LO}_\xi\big[\ell_\Delta(\mu_F)\big]-\log\frac{\vee}{k^+}\,H_4,}\qquad \ell_\Delta(\mu_F)=\log\frac{(y'-y)^2\mu_F^2}{c_0^2}.\end{aligned}\tag{5.1}$$

- The bracket in ξ is the plus distribution $\frac{1}{2N_c}P^{(r)}_{gg,+}$ on $[\frac{k^+}{\vee},1]$; its last term is your $\log(1-\frac{k^+}{\vee})$ line.
- $\mu_F^2\frac{\partial}{\partial\mu_F^2}D_{h/g}=\frac{\alpha_s}{2\pi}P_{gg}\otimes D_{h/g}$ cancels the μ_F dependence of $\ell_\Delta(\mu_F)$ and of (4.5).
- The last two lines ($-\log\frac{\vee}{k^+}H_4$) remove the log $\frac{\vee}{k^+}$ that the ξ integral produces at $\xi\to\frac{k^+}{\vee}$ (it is in the BFKL terms); with them the DGLAP term is finite for $\vee\to\infty$.
- Line 3 of the row (linear polarization) is finite and not DGLAP; it is in Sec. 6.

The pole that is left (as before, Q1). The second term of the pole, with its finite partner from Row II part 1:

$$\left.\frac{d^3N}{d^3k}\right|_{\rm Q1}=\frac{\alpha_sN_c}{4\pi}\log\Big(1-\frac{k^+}{\vee}\Big)\,{\rm LO}\big[\hat{E}_X+\hat{E}_{X'}\big].\tag{5.2}$$

It is not DGLAP (it is not $P_{gg}\otimes$LO) and not running coupling. It comes from the limit $\xi\geq\frac{k^+}{\vee}$ in the virtual term of Row II part 1 and vanishes for $k^+\ll\vee$, i.e. at mid rapidity in the high-energy limit. In AABKL the virtual term has $\int_0^1 d\xi$ and there is no such term (Sec. 11).

6 The finite terms

These are your rows with C1–C5, without the running coupling (Sec. 4) and the DGLAP term (Sec. 5). Blue marks a change with respect to your file. Three rows are reduced:

- Group I Row II part 1 keeps only $V_\vee$.
- Group I Row V 2b UV is gone.
- Group II Row IV keeps only line 3.

Every term is finite for $\epsilon\to0$, $\vee\to\infty$, $\Lambda\to0$; the terms with $\log(1\pm\frac{\Lambda}{k^+})$ vanish for $\Lambda\to0$. Group III is also written in the variables of AABKL in Sec. 11.

7 Finite terms: Group I (with c.c.)

7.1 Row I

NA

7.2 Row II

$$\begin{aligned}\frac{d^3N_{1\,\rm NLO}}{d^3k}&=\frac{1}{(2\pi)^3}\,\frac{g^4\,N_c}{32\pi^4}\,\frac{1}{k^+}\int_{w,w',x,x',}e^{-ik(w'-w)}\,\frac{(x'-w')\cdot(x-w)}{(x'-w')^2\,(x-w)^2}\left[\log^2\Big[1-\frac{k^+}{\vee}\Big]+\log\Big[1-\frac{k^+}{\vee}\Big]\,\log\Big[\frac{\vee}{k^+}\Big]\right]\times\\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\big[U^{ab}(w)\,\rho^b(x)-U^{ab}(x)\,\rho^b(x)\big]\end{aligned}\tag{7.1}$$

$$\frac{d^3N_{2\,b}}{d^3k} = -\frac{1}{(2\pi)^3}\,\frac{g^4}{64\pi^6}\,\frac{1}{k^+}\int_{w,w'} e^{-ik\cdot(w'-w)}\int d^2p\,d^2k\int_{x,y} e^{ik\cdot(w-y)}\,e^{ip\cdot(y-x)}\,\frac{(x'-w')^i}{(x'-w')^2\,\mathbf{p}^2\mathbf{k}^2}\,R^i(\mathbf{k},\mathbf{p})$$
$$\times\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\Big[f^{hbc}\,U^{ah}(w)\,\{\rho^b(x),\rho^c(y)\} - f^{abc}\,U^{bd}(x)\,U^{ce}(y)\,\{\rho^d(x),\rho^e(y)\}\Big] \tag{7.2}$$

where

$$R^i = -\,2k^i\log\left(\frac{p^2}{k^2+\frac{k^+}{\vee}\,p^2}\right) + \frac{(2p^2-2\,k\cdot p+k^2)\,k^i+k^2p^i}{(k-p)^2}\,\log\left[\frac{k^2}{(k-p)^2}+\frac{k^+}{\vee}\left(1-\frac{k^2}{(k-p)^2}\right)\right] - \frac{p^2k^i+k^2p^i}{(k-p)^2}\,\log\left(1-\frac{k^+}{\vee}\right). \tag{7.3}$$

Part 1: only $V_\vee = -\frac{\alpha_s}{2\pi}N_c\log(1-\frac{k^+}{\vee})\log(\frac{\vee}{k^+}-1)$ LO (with c.c.) is left; it vanishes for $k^+ \ll \vee$. The $\frac{11}{3}$ cancels against Row V 2b UV, the constant is in Eq. (4.6), the pole bracket times $\log(1-\frac{k^+}{\vee})$ is Q1, Eq. (5.2). Part 2b: no change.

7.3 Row III

$$\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\Bigg\{$$
$$-\,\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\,\log\left[1-\frac{k^+}{\vee}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{\big[(x-z)^2+(x-w)^2\big]}{\big[(x-z)^2-(x-w)^2\big]^2}$$
$$\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]$$
$$+\,\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{\big[(x-w)^2+(x-z)^2\big]}{\big[(x-z)^2-(x-w)^2\big]^2}\times\log\left[\frac{\Big[(x-z)^2-\frac{k^+}{\vee}\,(x-w)^2\Big]}{\big[\big(1+\frac{\Lambda}{k^+}\big)\,(x-z)^2-(x-w)^2\big]}\right]$$
$$\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]$$
$$-\,\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\left(-\log\left[1-\frac{\Lambda}{k^+}\right]\right)\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{\big[(x-z)^2+(x-w)^2\big]}{\big[(x-z)^2-(x-w)^2\big]^2}$$
$$\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]$$
$$+\,\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{\big[(x-w)^2+(x-z)^2\big]}{\big[(x-z)^2-(x-w)^2\big]^2}\times\log\left[\frac{\big[(k^+-\Lambda)\,(x-z)^2-k^+\,(x-w)^2\big]}{\big[\Lambda\,(x-z)^2-k^+\,(x-w)^2\big]}\right]$$
$$\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]$$
$$-\,\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\frac{(x'-w')\cdot(x-w)\,(x-z)\cdot(y-z)}{(x'-w')^2\,(x-w)^2\,(x-z)^2\,(y-z)^2}\,\log\left[\frac{\Big[\frac{k^+}{\vee}\,(x-w)^2-(x-z)^2\Big]}{\big[(x-w)^2-\frac{\Lambda}{k^+}\,(x-z)^2\big]}\right]$$
$$\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \tag{7.4}$$

$$\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = -\,\frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\Bigg\{$$
$$-\,\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\left(-\log\left[1+\frac{\Lambda}{k^+}\right]\right)\int_{x,y,z,x'}\frac{(x'-w')\cdot(x-w)\,(y-z)\cdot(z-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,(x-w)^2}$$
$$\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]$$
$$+\,\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left[1-\frac{k^+}{\vee}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\big[(x-w)^2-(x-z)^2\big]^2}$$
$$\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]$$
$$-\,\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left[1-\frac{k^+}{\vee}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\big[(x-w)^2-(x-z)^2\big]^2}$$
$$\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]$$
$$+\,\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left[\frac{(x-w)^2\,\frac{k^+}{\vee}-(x-z)^2}{(x-w)^2-(x-z)^2\,\big(1+\frac{\Lambda}{k^+}\big)}\right]\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\big[(x-w)^2-(x-z)^2\big]}$$
$$\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]$$

$$
\begin{aligned}
&- \frac{ig^4\, f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'} \log\left[\frac{(x-w)^2\,\frac{k^+}{\vee}-(x-z)^2}{(x-w)^2-(x-z)^2\,\left(1+\frac{\Lambda}{k^+}\right)}\right]\frac{(x'-w')\cdot(x-w)\,(y-z)\cdot(z-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]\,(x-w)^2}\\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]\\
&+ \frac{ig^4\, f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'} \log\left[\frac{(x-w)^2\,\frac{k^+}{\vee}-(x-z)^2}{(x-w)^2-(x-z)^2\,\left(1+\frac{\Lambda}{k^+}\right)}\right]\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2}\\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]\\
&- \frac{ig^4\, f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'} \log\left[\frac{(x-w)^2\,\frac{k^+}{\vee}-(x-z)^2}{(x-w)^2-(x-z)^2\,\left(1+\frac{\Lambda}{k^+}\right)}\right]\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]\,(x-z)^2}\\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]\\
&+ \frac{ig^4\, f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'} \log\left[\frac{(x-w)^2\,\frac{k^+}{\vee}-(x-z)^2}{(x-w)^2-(x-z)^2\,\left(1+\frac{\Lambda}{k^+}\right)}\right]\frac{(x'-w')\cdot(x-z)\,(y-z)\cdot(z-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]}\\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]\\
&- \frac{ig^4\, f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'} \log\left[\frac{(x-w)^2\,\frac{k^+}{\vee}-(x-z)^2}{(x-w)^2-(x-z)^2\,\left(1+\frac{\Lambda}{k^+}\right)}\right]\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2}\\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]\Bigg\}
\end{aligned}
\tag{7.5}
$$

$$
\begin{aligned}
\frac{d^3N_{3a\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\Bigg\{\\
&- \frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left[1-\frac{\Lambda}{k^+}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2(y-z)^2(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2}\\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]\\
&+ \frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left[1-\frac{\Lambda}{k^+}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2(y-z)^2(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2}\\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]\\
&- \frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{(x-w)^2\,k^+-(x-z)^2\,(k^+-\Lambda)}{(x-w)^2\,k^+-(x-z)^2\,\Lambda}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2(y-z)^2(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2}\\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]\\
&+ \frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{(x-w)^2\,k^+-(x-z)^2\,(k^+-\Lambda)}{(x-w)^2\,k^+-(x-z)^2\,\Lambda}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2(y-z)^2(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2}\\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]\\
&- \frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{(x-w)^2\,k^+-(x-z)^2\,(k^+-\Lambda)}{(x-w)^2\,k^+-(x-z)^2\,\Lambda}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-w)}{(x'-w')^2(y-z)^2(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]}\\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]\\
&+ \frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{(x-w)^2\,k^+-(x-z)^2\,(k^+-\Lambda)}{(x-w)^2\,k^+-(x-z)^2\,\Lambda}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)\,(x-w)^2}{(x'-w')^2(y-z)^2(z-w)^2\,\left[(x-w)^2-(x-z)^2\right](x-z)^2}\\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]\\
&+ \frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{(x-w)^2\,k^+-(x-z)^2\,(k^+-\Lambda)}{(x-w)^2\,k^+-(x-z)^2\,\Lambda}\right]\int_{x,y,z,x'}\frac{(y-z)\cdot(z-w)\,(x'-w')\cdot(x-w)\,(x-z)^2}{(x'-w')^2(y-z)^2(z-w)^2\,\left[(x-w)^2-(x-z)^2\right](x-w)^2}\\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]\\
&- \frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{(x-w)^2\,k^+-(x-z)^2\,(k^+-\Lambda)}{(x-w)^2\,k^+-(x-z)^2\,\Lambda}\right]\int_{x,y,z,x'}\frac{(y-z)\cdot(z-w)\,(x'-w')\cdot(x-z)}{(x'-w')^2(y-z)^2(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]}\\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]\\
&+ \frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\mathcal{A}\int_{x,y,z,x'}\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-w)}{(x'-w')^2(y-z)^2(z-w)^2}
\end{aligned}
$$

$$\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]$$
$$+ \frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\,\mathcal{A}\int_{x,y,z,x'} \frac{(y-z)\cdot(z-w)\,(x'-w')\cdot(x-z)}{(x'-w')^2(y-z)^2(z-w)^2}$$
$$\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]$$
$$- \frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\,\mathcal{A}\int_{x,y,z,x'} \frac{(x-w)\cdot(x-z)\,(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)}{(x'-w')^2(y-z)^2(z-w)^2(x-z)^2}$$
$$\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]\Bigg\} \tag{7.6}$$

$$\frac{d^3N_{3b\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\Bigg\{$$
$$- \frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\,\mathcal{A}\int_{x,y,z,x'} \frac{(x-w)\cdot(x-z)\,(y-z)\cdot(z-w)\,(x'-w')\cdot(x-w)}{(x'-w')^2(y-z)^2(z-w)^2(x-w)^2}$$
$$\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]$$
$$- \frac{ig^4 f^{abd}}{16\pi^5}\frac{1}{k^+}\,\log\left[\frac{\left[k^+(x-w) - (k^+-\Lambda)(x-z)\right]^2}{\left[k^+(x-w) - \Lambda(x-z)\right]^2}\right]\int_{x,y,z,x'} \frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)}{(x'-w')^2(y-z)^2(z-w)^2(x-z)^2}$$
$$\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]$$
$$+ \frac{ig^4 f^{abd}}{16\pi^5}\frac{1}{k^+}\,\log\left[\frac{\left[k^+(x-w) - (k^+-\Lambda)(x-z)\right]^2}{\left[k^+(x-w) - \Lambda(x-z)\right]^2}\right]\int_{x,y,z,x'} \frac{(y-z)\cdot(z-w)\,(x'-w')\cdot(x-w)}{(x'-w')^2(y-z)^2(z-w)^2(x-w)^2}$$
$$\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]\Bigg\} \tag{7.7}$$

where

$$\mathcal{A} = \frac{1}{\sqrt{(x-w)^2(x-z)^2 - [(x-w)\cdot(x-z)]^2}}\Bigg\{\arctan\left[\frac{(k^+-\Lambda)(x-z)^2 - k^+(x-w)\cdot(x-z)}{k^+\sqrt{(x-w)^2(x-z)^2 - [(x-w)\cdot(x-z)]^2}}\right]$$
$$- \arctan\left[\frac{\Lambda(x-z)^2 - k^+(x-w)\cdot(x-z)}{k^+\sqrt{(x-w)^2(x-z)^2 - [(x-w)\cdot(x-z)]^2}}\right]\Bigg\}$$

The edge-kernel terms are removed (C4). Their place is taken by the finite part of the $\mathcal{D}$ region (Q3, open).

7.4 Row IV

$$\frac{d^3N_{\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'} e^{-ik\cdot(w'-w)}\Bigg\{$$
$$\frac{ig^4}{8\pi^5}\frac{1}{k^+}\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\times\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2 + \frac{k^+}{\vee}(x-w)^2}{\frac{\Lambda}{k^+}(x-z)^2 + (x-w)^2}\right]\times\left[\frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}\right.$$
$$\left. - \frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2} - \frac{(x-z)^j\,(x-w)^i}{(x-z)^2\,(x-w)^2} + \frac{\delta^{ij}}{2\,(z-w)^2}\right]$$
$$- \frac{ig^4}{16\pi^5}\frac{1}{k^+}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2\,(z-w)^2}\times\left(-\log\left[1+\frac{\Lambda}{k^+}\right]\right)$$
$$+ \frac{ig^4}{8\pi^5}\frac{1}{k^+}\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\times\log\left[1-\frac{k^+}{\vee}\right]\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2} + \frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\right]\Bigg\}\times$$
$$\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\Big[f^{cdc'}\,U^{ac}(w)\,U^{bd}(z)\,U^{be}(y)\,\rho^{e}(y)\rho^{c'}(x) - f^{abc}\,U^{be}(y)\,U^{cd}(x)\,\rho^{e}(y)\rho^{d}(x)\Big] \tag{7.8}$$

7.5 Row V

First part

$$\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\frac{ig^4}{16\pi^4}\frac{1}{k^{+2}}\int_0^{k^+}\frac{dp^+}{2\pi}\int_{x,z,y,x',y'w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\frac{p^+}{k^+}}\,\frac{(x'-w')^i\,(z-x)^m\,(y-z)^k\,(y'-x)^j}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2}$$

$$
\left[\delta_{im}\,\delta_{jk}-\left[\frac{k^+}{p^+}\right]_+\delta_{jm}\,\delta_{ik}\right]\times\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times
$$

$$
\left[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^e(y),\rho^b(y')\}-f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^d(y),\rho^e(y')\}\right]
$$

$$
-\frac{1}{(2\pi)^3}\,\frac{ig^4}{16\pi^4}\,\frac{1}{k^+}\int_0^{k^+}\frac{dq^+}{2\pi}\int_{x,z,y,x',y'w'}e^{-ik\cdot(w'-x)}\,e^{-ik\cdot(x-z)\frac{q^+}{k^+}}\,\frac{(x'-w')\cdot(y'-x)\,(z-x)\cdot(y-z)}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2}\,\left[\frac{1}{q^+}\right]_+
$$

$$
\times\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^e(y),\rho^b(y')\}-f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^d(y),\rho^e(y')\}\right]
$$

$$
-\frac{1}{(2\pi)^3}\int_{x,z,y,x',y'w'}e^{-ik\cdot(w'-z)}\,\frac{ig^4}{32\pi^5}\,\frac{1}{k^+}\,\log\left[1-\frac{\Lambda}{k^+}\right]\,\frac{(x'-w')\cdot(y-z)\,(z-x)\cdot(y'-x)}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2}
$$

$$
\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^e(y),\rho^b(y')\}-f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^d(y),\rho^e(y')\}\right]
$$

$$
-\frac{1}{(2\pi)^3}\int_{x,z,y,x',y'w'}e^{-ik\cdot(w'-x)}\,\frac{ig^4}{32\pi^5}\,\frac{1}{k^+}\,\log\left[1-\frac{\Lambda}{k^+}\right]\,\frac{(x'-w')\cdot(y'-x)\,(z-x)\cdot(y-z)}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2}
$$

$$
\times\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^e(y),\rho^b(y')\}-f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^d(y),\rho^e(y')\}\right] \tag{7.9}
$$

Second part

$$
\frac{d^3N_{2a\,\mathrm{NLO}}}{d^3k}=\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_0^{k^+}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'}e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\,\frac{p^+}{k^+}}\,\frac{1}{[p^+\,(y-x)^2+(k^+-p^+)\,(y-z)^2]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}
$$

$$
\left[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\,\left[(y-x)^2-(y-z)^2\right]\,\frac{p^+\,(k^+-p^+)}{k^{+\,2}}\right.
$$

$$
+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\,\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2\,(y-x)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]
$$

$$
-\left[\frac{k^+}{p^+}\right]_+\,\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]+\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]
$$

$$
+\frac{p^+\,\delta_{im}}{k^+}\,\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2\,(y-z)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]
$$

$$
\left.-\left[\frac{(y-x)^m\,(x-z)^i}{(x-z)^2}-\frac{(y-x)^m\,(y-z)^i}{2\,(y-z)^2}+\frac{(y-z)^m\,(x-z)^i}{(x-z)^2}+\frac{(y-x)^i\,(y-z)^m}{2\,(y-x)^2}\right]\right]\times
$$

$$
\left[U^{ab'}(x')-U^{ab'}(w')\right]\,\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(x)\right]\,\rho^{b'}(x')\,\rho^f(y)
$$

$$
+\frac{1}{(2\pi)^3}\int_{k^+}^0\frac{dq^+}{2\pi}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{x,y,z,x',w'}e^{-ik\cdot(w'-x)}\,e^{-ik\cdot(x-z)\,\frac{q^+}{k^+}}\,\frac{1}{[(k^+-q^+)\,(y-x)^2+q^+\,(y-z)^2]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}
$$

$$
\left\{\left[\frac{k^+}{q^+}\right]_+\,\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]-\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]\right\}\times
$$

$$
\left[U^{ab'}(x')-U^{ab'}(w')\right]\,\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(x)\right]\,\rho^{b'}(x')\,\rho^f(y)
$$

$$
-\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[1-\frac{\Lambda}{k^+}\right]\,\int_{x,y,z,x',w'}e^{-ik\cdot(w'-z)}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2\,(y-z)^2}\,\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}\right.
$$

$$
\left.+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]\times\left[U^{ab'}(x')-U^{ab'}(w')\right]\,\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(x)\right]\,\rho^{b'}(x')\,\rho^f(y)
$$

$$
-\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[1-\frac{\Lambda}{k^+}\right]\,\int_{x,y,z,x',w'}e^{-ik\cdot(w'-x)}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2\,(y-x)^2}
$$

$$
\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]\,\left[U^{ab'}(x')-U^{ab'}(w')\right]\,\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(x)\right]\,\rho^{b'}(x')\,\rho^f(y) \tag{7.10}
$$

$d^3N_{2b\,\mathrm{UV}}$: nothing is left. Its $-\frac{11}{6}$ cancels the $\frac{11}{3}$ of Row II part 1 (Sec. 4); $2\log(1-\frac{\Lambda}{k^+})\to 0$.

$$
\begin{aligned}
\frac{d^3N_{2b\,\text{finite NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_\Lambda^{k^+-\Lambda}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\frac{p^+}{k^+}}\,\frac{1}{[p^+(y-x)^2+(k^+-p^+)(y-z)^2]}\\
&\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\times\left[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\left[(y-x)^2-(y-z)^2\right]\frac{p^+(k^+-p^+)}{k^{+2}}\right.\\
&+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2(y-x)^2}-\frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\right]\\
&-\left[\frac{k^+}{p^+}\right]_+\left[\frac{(y-z)^i(x-z)^m}{(x-z)^2}+\frac{(y-x)^m(y-z)^i}{2(y-x)^2}\right]+\left[\frac{(y-z)^i(x-z)^m}{(x-z)^2}+\frac{(y-x)^m(y-z)^i}{2(y-x)^2}\right]\\
&+\frac{p^+\,\delta_{im}}{k^+}\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2(y-z)^2}-\frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\right]\\
&\left.-\left[\frac{(y-x)^m(x-z)^i}{(x-z)^2}-\frac{(y-x)^m(y-z)^i}{2(y-z)^2}+\frac{(y-z)^m(x-z)^i}{(x-z)^2}+\frac{(y-x)^i(y-z)^m}{2(y-x)^2}\right]\right]\\
&\times\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y)-N_c\,U^{af}(x)\Big]\rho^{b'}(x')\,\rho^f(y)\\
&+\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{k^+}^0\frac{dq^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\;e^{-ik\cdot(z-x)\frac{(k^+-q^+)}{k^+}}\,\frac{1}{[(k^+-q^+)(y-x)^2+q^+\,(y-z)^2]}\\
&\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\left[\frac{(y-x)^i(x-z)^m}{(x-z)^2}-\frac{(y-x)^i(y-z)^m}{2(y-z)^2}\right]\left[\frac{k^+}{q^+}\right]_+\times\\
&\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y)-N_c\,U^{af}(x)\Big]\rho^{b'}(x')\,\rho^f(y)\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{k^+-\Lambda}^{\Lambda}\frac{dq^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\;e^{-ik\cdot(z-x)\frac{(k^+-q^+)}{k^+}}\,\frac{1}{[(k^+-q^+)(y-x)^2+q^+\,(y-z)^2]}\\
&\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\left[\frac{(y-x)^i(x-z)^m}{(x-z)^2}-\frac{(y-x)^i(y-z)^m}{2(y-z)^2}\right]\\
&\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y)-N_c\,U^{af}(x)\Big]\rho^{b'}(x')\,\rho^f(y)\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[1-\frac{\Lambda}{k^+}\right]\int_{x,y,z,x',w'} e^{-ik\cdot(w'-x)}\left[\frac{(y-x)\cdot(x'-w')\,(x-z)\cdot(z-x)}{(x-z)^2\,(x'-w')^2\,(z-x)^2\,(y-x)^2}\right.\\
&\left.-\frac{(y-x)\cdot(x'-w')\,(y-z)\cdot(z-x)}{2(y-z)^2\,(x'-w')^2\,(z-x)^2\,(y-x)^2}\right]\times\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y)-N_c\,U^{af}(x)\Big]\rho^{b'}(x')\,\rho^f(y)\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[1-\frac{\Lambda}{k^+}\right]\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\left[\frac{(y-z)\cdot(x'-w')\,(x-z)\cdot(z-x)}{(x-z)^2\,(x'-w')^2\,(z-x)^2\,(y-z)^2}\right.\\
&\left.+\frac{(y-x)\cdot(z-x)\,(y-z)\cdot(x'-w')}{2(y-x)^2\,(x'-w')^2\,(z-x)^2\,(y-z)^2}\right]\times\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y)-N_c\,U^{af}(x)\Big]\rho^{b'}(x')\,\rho^f(y)\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\left[-\frac{11}{6}+2\,\log\left[1-\frac{\Lambda}{k^+}\right]\right]\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{(y-x)\cdot(y-z)}{(y-x)^2\,(x-z)^2}\times\\
&\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y)-N_c\,U^{af}(z)\Big]\rho^{b'}(x')\,\rho^f(y)
\end{aligned}
\tag{7.11}
$$

8 Finite terms: Group II

8.1 Row I

8.1.1 First part

$$
\begin{aligned}
\frac{d^3N_{1\,\text{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'} e^{-ik(w'-w)}\Bigg\{\\
&-\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\,\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{\frac{\Lambda}{k^+}\,(x-z)^2+(x-w)^2}\right]\times\\
&\left[\frac{\delta^{ij}}{2\,(z-w)^2}+\frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}-\frac{(x-z)^j\,(x-w)^i}{(x-w)^2\,(x-z)^2}-\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}\right]\\
&+\frac{ig^4}{32\pi^5}\,\frac{1}{k^+}\,\left(-\log\left[1+\frac{\Lambda}{k^+}\right]\right)\,\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2\,(z-w)^2}
\end{aligned}
$$

$$
- \frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[1-\frac{k^+}{\vee}\right]\,\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\,\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}+\frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\right]\Bigg\}
$$

$$
\left[U^{ac'}(w')\,\{\rho^{c'}(x'),\rho^b(y)\}-U^{bc'}(z)\,U^{ad'}(x')\,U^{c'e'}(y)\,\{\rho^{d'}(x'),\rho^{e'}(y)\}\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right]
$$

$$
+\frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'}e^{-ik(w'-w)}\Bigg\{
$$

$$
\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w)^i\,(y-z)^j}{(x'-w)^2\,(y-z)^2}\,\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w')^2}{\frac{\Lambda}{k^+}\,(x-z)^2+(x-w')^2}\right]\times
$$

$$
\left[\frac{\delta^{ij}}{2\,(z-w')^2}+\frac{(x-z)^j\,(z-w')^i}{(z-w')^2\,(x-z)^2}-\frac{(x-z)^j\,(x-w')^i}{(x-w')^2\,(x-z)^2}-\frac{(x-w')^i\,(z-w')^j}{(z-w')^2\,(x-w')^2}\right]
$$

$$
-\frac{ig^4}{32\pi^5}\,\frac{1}{k^+}\,\left(-\log\left[1+\frac{\Lambda}{k^+}\right]\right)\,\frac{(x'-w)\cdot(y-z)}{(x'-w)^2\,(y-z)^2\,(z-w')^2}
$$

$$
+\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[1-\frac{k^+}{\vee}\right]\,\frac{(x'-w)^i\,(y-z)^j}{(x'-w)^2\,(y-z)^2}\,\left[\frac{(x-w')^i\,(z-w')^j}{(z-w')^2\,(x-w')^2}+\frac{(x-z)^j\,(x-w')^i}{2\,(x-z)^2\,(x-w')^2}\right]\Bigg\}
$$

$$
\times\left[f^{c'bd'}\,U^{ac'}(w')\,\rho^{d'}(x)-f^{ac'd'}\,U^{bc'}(z)\,U^{d'e'}(x)\,\rho^{e'}(x)\right]\times\left[U^{ac}(w)\,\{\rho^c(x'),\rho^b(y)\}-U^{bc}(z)\,U^{ad}(x')\,U^{ce}(y)\,\{\rho^d(x'),\rho^e(y)\}\right] \tag{8.1}
$$

8.1.2 Second part

$$
\begin{aligned}
\frac{d^3N_{2a\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'}e^{-ik(w'-w)}\Bigg\{\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}\times\\
&\left[\frac{(x-w)\cdot(z-w')\,(x'-z)\cdot(z-w)}{(z-w)^2\,(z-w')^2}-\frac{(x-w)\cdot(x'-w')\,(z-w)\cdot(x'-z)}{2\,(z-w)^2\,(x'-w')^2}\right.\\
&+\frac{(x-z)\cdot(x'-z)\,(x-w)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-w')^2}\\
&+\frac{(x'-w')\cdot(z-w)\,(x-z)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(z-w)\,(x-z)\cdot(x'-z)}{2\,(z-w)^2\,(x'-z)^2}\\
&\left.-\frac{(x-z)\cdot(z-w')\,(x'-w')\cdot(x-w)}{2\,(x-w)^2\,(z-w')^2}-\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(x-z)}{4\,(x-w)^2\,(x'-z)^2}\right]\times\\
&\left[\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{\frac{k^+}{\vee}\,(x'-w')^2+\left(1-\frac{k^+}{\vee}\right)(x'-z)^2}\right]-\log\left[\frac{\Lambda\,(x-z)^2+k^+\,(x-w)^2}{k^+\,(x'-w')^2+\Lambda\,(x'-z)^2}\right]\right]\Bigg\}\times\\
&\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^d(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^e(x)\right.\\
&\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^d(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^e(x)\right]
\end{aligned} \tag{8.2}
$$

$$
\begin{aligned}
\frac{d^3N_{2b\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\left[1-\frac{k^+}{\vee}\right]\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{1}{[(x'-w')^2\,(x-z)^2\,(x-w)^2-(x-w)^4\,(x'-z)^2]}\times\\
&\left[\frac{(x'-w')\cdot(x-w)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(z-w)}{2\,(z-w)^2\,(x'-z)^2}\right.\\
&\left.+\frac{(x-w)\cdot(x'-w')\,(x-z)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-z)^2}\right]\times\left[(x-z)^2-\frac{(x-w)^2\,(x'-z)^2}{(x'-w')^2}\right]\times\\
&\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^d(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^e(x)\right.\\
&\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^d(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^e(x)\right]\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{(x-z)^2}{[(x'-w')^2\,(x-z)^2\,(x-w)^2-(x-w)^4\,(x'-z)^2]}\times
\end{aligned}
$$

$$
\left[\frac{(x'-w')\cdot(x-w)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(z-w)}{2\,(z-w)^2\,(x'-z)^2}\right.
$$

$$
\left.+\frac{(x-w)\cdot(x'-w')\,(x-z)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-z)^2}\right]
$$

$$
\times\,\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{\frac{\Lambda}{k^+}\,(x-z)^2+(x-w)^2}\right]\times
$$

$$
\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.
$$

$$
\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right]
$$

$$
+\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{(x'-z)^2}{[(x'-w')^4\,(x-z)^2\;-(x-w)^2\,(x'-z)^2\,(x'-w')^2]}\times
$$

$$
\left[\frac{(x'-w')\cdot(x-w)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(z-w)}{2\,(z-w)^2\,(x'-z)^2}\right.
$$

$$
\left.+\frac{(x-w)\cdot(x'-w')\,(x-z)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-z)^2}\right]
$$

$$
\times\,\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x'-z)^2+\frac{k^+}{\vee}\,(x'-w')^2}{\frac{\Lambda}{k^+}\,(x'-z)^2+(x'-w')^2}\right]
$$

$$
\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.
$$

$$
\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right] \tag{8.3}
$$

$$
\frac{d^3N_{2c\,\mathrm{NLO}}}{d^3k}=\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{(x'-w')^2}{(x'-z)^2}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}
$$

$$
\left[\frac{(x'-z)\cdot(x-z)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x-z)\,(x'-w')\cdot(z-w)}{2\,(z-w)^2\,(x'-w')^2}-\frac{(x-w)\cdot(z-w')\,(x-z)\cdot(x'-z)}{2\,(x-w)^2\,(z-w')^2}\right.
$$

$$
\left.+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-w)^2\,(x'-w')^2}\right]\times\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x'-z)^2+\frac{k^+}{\vee}\,(x'-w')^2}{\frac{\Lambda}{k^+}\,(x'-z)^2+(x'-w')^2}\right]
$$

$$
\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.
$$

$$
\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right]
$$

$$
-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{(x-w)^2}{(x-z)^2}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}
$$

$$
\left[\frac{(x'-z)\cdot(x-z)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x-z)\,(x'-w')\cdot(z-w)}{2\,(z-w)^2\,(x'-w')^2}-\frac{(x-w)\cdot(z-w')\,(x-z)\cdot(x'-z)}{2\,(x-w)^2\,(z-w')^2}\right.
$$

$$
\left.+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-w)^2\,(x'-w')^2}\right]\times\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{\frac{\Lambda}{k^+}\,(x-z)^2+(x-w)^2}\right]
$$

$$
\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.
$$

$$
\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right] \tag{8.4}
$$

$$
\frac{d^3N_{2d\,\mathrm{NLO}}}{d^3k}=-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\left(-\log\left[1+\frac{\Lambda}{k^+}\right]\right)\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{1}{[(x-z)^2-(x-w)^2]}\,\frac{1}{[(x'-w')^2-(x'-z)^2]}
$$

$$
\left[\frac{(x-w)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-z)^2\,(z-w')^2}\right.
$$

$$
\left.+\frac{(x'-w')\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-z)^2}\right]
$$

$$
\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.
$$

$$
\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right]
$$

$$
+\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{(x-z)^2}{[(x-z)^2-(x-w)^2]}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}
$$

$$
\left[\frac{(x-w)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-z)^2\,(z-w')^2}\right.
$$

$$
\left.+\frac{(x'-w')\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-z)^2}\right]\times
$$

$$
\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{\frac{\Lambda}{k^+}\,(x-z)^2+(x-w)^2}\right]\times
$$

$$
\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.
$$

$$
\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right]
$$

$$
+\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{(x'-z)^2}{[(x'-w')^2-(x'-z)^2]}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}
$$

$$
\left[\frac{(x-w)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-z)^2\,(z-w')^2}\right.
$$

$$
\left.+\frac{(x'-w')\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-z)^2}\right]\times
$$

$$
\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x'-z)^2+\frac{k^+}{\vee}\,(x'-w')^2}{\frac{\Lambda}{k^+}\,(x'-z)^2+(x'-w')^2}\right]\times
$$

$$
\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.
$$

$$
\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right] \tag{8.5}
$$

$$
\frac{d^3N_{2e\,\mathrm{NLO}}}{d^3k}=\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\left(-\log\left[1+\frac{\Lambda}{k^+}\right]\right)\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{1}{[(x-z)^2-(x-w)^2]}\,\frac{1}{[(x'-w')^2-(x'-z)^2]}
$$

$$
\left[\frac{(x-z)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-w)^2\,(z-w')^2}\right.
$$

$$
\left.+\frac{(x'-z)\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-w')^2}\right]
$$

$$
\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.
$$

$$
\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right]
$$

$$
-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{(x-w)^2}{[(x-z)^2-(x-w)^2]}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}
$$

$$
\left[\frac{(x-z)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-w)^2\,(z-w')^2}\right.
$$

$$
+\frac{(x'-z)\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-w')^2}\Bigg]\times
$$

$$
\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{\frac{\Lambda}{k^+}\,(x-z)^2+(x-w)^2}\right]\times
$$

$$
\Bigg[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)
$$

$$
-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Bigg]
$$

$$
-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{(x'-w')^2}{\left[(x'-w')^2-(x'-z)^2\right]}\,\frac{1}{\left[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2\right]}
$$

$$
\Bigg[\frac{(x-z)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-w)^2\,(z-w')^2}
$$

$$
+\frac{(x'-z)\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-w')^2}\Bigg]\times
$$

$$
\log\left[\frac{\frac{k^+}{\vee}\,(x'-w')^2+\left(1-\frac{k^+}{\vee}\right)(x'-z)^2}{(x'-w')^2+\frac{\Lambda}{k^+}\,(x'-z)^2}\right]\times
$$

$$
\Bigg[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)
$$

$$
-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Bigg] \tag{8.6}
$$

$$
\frac{d^3N_{2f\,\mathrm{NLO}}}{d^3k}=\frac{1}{(2\pi)^3}\,\frac{g^4\,d_\perp}{32\pi^5}\,\left(-\log\left[1+\frac{\Lambda}{k^+}\right]\right)\,\frac{1}{k^+}\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{\left[(x'-w')^2\,(x-w)^2-(x-z)^2\,(x'-z)^2\right]}{(z-w)^2\,(z-w')^2\,\left[(x-z)^2-(x-w)^2\right]\,\left[(x'-z)^2-(x'-w'\right.}
$$

$$
\left[f^{c'bd'}\,U^{ac'}(w')\,\rho^{d'}(x')-f^{ac'd'}\,U^{bc'}(z)U^{e'd'}(x')\,\rho^{e'}(x')\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^{d}(x)-f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^{e}(x)\right]
$$

$$
-\frac{1}{(2\pi)^3}\,\frac{g^4\,d_\perp}{32\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{(x-z)^2\,(x-w)^2}{(z-w)^2\,(z-w')^2\,\left[(x-z)^2-(x-w)^2\right]}\,\frac{\left[(x'-z)^2-(x'-w')^2\right]}{\left[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2\right]}
$$

$$
\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{\frac{\Lambda}{k^+}\,(x-z)^2+(x-w)^2}\right]\times
$$

$$
\Bigg[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)
$$

$$
-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Bigg]
$$

$$
+\frac{1}{(2\pi)^3}\,\frac{g^4\,d_\perp}{32\pi^5}\,\left[\frac{1}{\vee}-\frac{1}{k^++\Lambda}\right]\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{1}{(z-w)^2\,(z-w')^2}
$$

$$
\Bigg[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)
$$

$$
-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Bigg]
$$

$$
+\frac{1}{(2\pi)^3}\,\frac{g^4\,d_\perp}{32\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{(x'-w')^2\,(x'-z)^2}{(z-w)^2\,(z-w')^2}
$$

$$
\frac{\left[(x-z)^2-(x-w)^2\right]}{\left[(x'-z)^2-(x'-w')^2\right]\,\left[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2\right]}
$$

$$
\times\,\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x'-z)^2+\frac{k^+}{\vee}\,(x'-w')^2}{\frac{\Lambda}{k^+}\,(x'-z)^2+(x'-w')^2}\right]\times
$$

$$
\Bigg[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)
$$

$$- f^{ac'd'} f^{cbd} U^{ac}(w) U^{bc'}(z) U^{e'd'}(x') \rho^{e'}(x') \rho^{d}(x) + N_c U^{e'd}(x') U^{de}(x) \rho^{e'}(x') \rho^{e}(x) \Bigg] \tag{8.7}$$

8.2 Row II

NA

8.3 Row III

NA

8.4 Row IV

$$\begin{aligned}\frac{d^3N_{\rm NLO}}{d^3k} = \frac{g^2}{4\pi^2}\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{N_c}{k^+}\int_{\frac{k^+}{\vee}}^{\frac{k^+}{k^++\Lambda}} d\xi \frac{(1-\xi)}{\xi}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}} \\ \times\left[\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2} - \frac{(x-y)\cdot(y'-y)(x'-y')\cdot(y'-y)}{(x-y)^2(x'-y')^2(y'-y)^2}\, d_\perp\right]\times \\ \left[U^{e'c}(y') - U^{e'c}(x')\right]\times\left[U^{ec}(y) - U^{ce}(x)\right]\rho^{e'}(x')\rho^{e}(x)\end{aligned} \tag{8.8}$$

Only line 3 (linear polarization) is left. Lines 1, 2 and $-\log\frac{\vee}{k^+}H_4$ (C1) are the DGLAP term, Eq. (5.1).

9 Finite terms: Group III (with c.c.)

9.1 Row I

$$\begin{aligned}\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\int_{x,y,x',z,w,w'} e^{-ik(w'-w)}\Bigg\{ \\ -\frac{ig^4}{8\pi^5}\frac{1}{k^+}\frac{(x'-w')^i(y-z)^j}{(x'-w')^2(y-z)^2}\times\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}(x-w)^2}{(x-w)^2+\frac{\Lambda}{k^+}(x-z)^2}\right]\times\left[\frac{\delta^{ij}}{2(z-w)^2}+\frac{(x-z)^j(z-w)^i}{(z-w)^2(x-z)^2}\right. \\ \left.-\frac{(x-z)^j(x-w)^i}{(x-w)^2(x-z)^2}-\frac{(x-w)^i(z-w)^j}{(z-w)^2(x-w)^2}\right] \\ +\frac{ig^4}{16\pi^5}\frac{1}{k^+}\left(-\log\left[1+\frac{\Lambda}{k^+}\right]\right)\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2(z-w)^2} \\ -\frac{ig^4}{8\pi^5}\frac{1}{k^+}\log\left[1-\frac{k^+}{\vee}\right]\frac{(x'-w')^i(y-z)^j}{(x'-w')^2(y-z)^2}\left[\frac{(x-w)^i(z-w)^j}{(z-w)^2(x-w)^2}+\frac{(x-z)^j(x-w)^i}{2(x-z)^2(x-w)^2}\right]\Bigg\}\times \\ \left[U^{ab'}(x')U^{c'd'}(y)U^{c'b}(z)\rho^{b'}(x')\rho^{d'}(y) - U^{bd'}(z)U^{d'e'}(y)U^{ac'}(w')\rho^{c'}(x')\rho^{e'}(y)\right]\times \\ \left[f^{cbd}U^{ac}(w)\rho^{d}(x) - f^{acd}U^{bc}(z)U^{de}(x)\rho^{e}(x)\right]\end{aligned} \tag{9.1}$$

9.2 Row II

$$\begin{aligned}\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\int_{x,y,x',z,w,w'} e^{-ik(w'-w)}\Bigg\{ \\ -\frac{ig^4}{8\pi^5}\frac{1}{k^+}\frac{(x'-w')^i(y-z)^j}{(x'-w')^2(y-z)^2}\times\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}(x-w)^2}{(x-w)^2+\frac{\Lambda}{k^+}(x-z)^2}\right]\times\left[\frac{\delta^{ij}}{2(z-w)^2}+\frac{(x-z)^j(z-w)^i}{(z-w)^2(x-z)^2}\right. \\ \left.-\frac{(x-w)^i(z-w)^j}{(z-w)^2(x-w)^2}-\frac{(x-z)^j(x-w)^i}{(x-z)^2(x-w)^2}\right] \\ +\frac{ig^4}{16\pi^5}\frac{1}{k^+}\left(-\log\left[1+\frac{\Lambda}{k^+}\right]\right)\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2(z-w)^2} \\ -\frac{ig^4}{8\pi^5}\frac{1}{k^+}\log\left[1-\frac{k^+}{\vee}\right]\frac{(x'-w')^i(y-z)^j}{(x'-w')^2(y-z)^2}\left[\frac{(x-w)^i(z-w)^j}{(z-w)^2(x-w)^2}+\frac{(x-z)^j(x-w)^i}{2(x-z)^2(x-w)^2}\right]\Bigg\}\times\end{aligned}$$

$$\left[U^{c'd'}(y)\,U^{ab'}(x')\,U^{bc'}(z)\,\rho^{d'}(y)\rho^{b'}(x') - U^{ab'}(x')\,\rho^{b}(y)\rho^{b'}(x')\right] \times \left[f^{cbd}\,U^{ac}(w)\,\rho^{d}(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^{e}(x)\right] \tag{9.2}$$

9.3 Row III

9.3.1 First part

$$\begin{aligned}
\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{8\pi^4}\,\frac{1}{k^+}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,x',y',z,y,x} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\\
&\left[\frac{\delta_{k'm}\,\delta_{ij}}{(p^+ + k^+)} - \left[\frac{1}{p^+}\right]_+^{(k^+)}\delta_{jm}\,\delta_{ik'} - \frac{1}{k^+}\,\delta_{im}\,\delta_{jk'}\right]\left[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\\
&\left[U^{ac}(w)\,\{\rho^c(y),\rho^b(x)\} - U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\{\rho^d(y),\rho^e(x)\}\right]\\
&- \frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{16\pi^5}\,\frac{1}{k^+}\,\log\left[1-\frac{k^+}{\vee}\right]\int_{w,x',y',z,y,x} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\\
&\left[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\left[U^{ac}(w)\,\{\rho^c(y),\rho^b(x)\} - U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\{\rho^d(y),\rho^e(x)\}\right]\\
&+ \frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{8\pi^4}\,\frac{1}{k^+}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\,\frac{1}{p^+ + k^+}\int_{w,x',y',z,y,x} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(y-w)\,(x'-z)\cdot(x-z)}{(y'-z)^2\,(y-w)^2\,(x'-z)^2\,(x-z)^2}\\
&\left[U^{e'c'}(z)\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\left[U^{ac}(w)\,\{\rho^c(y),\rho^b(x)\} - U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\{\rho^d(y),\rho^e(x)\}\right]
\end{aligned} \tag{9.3}$$

C2: $\left[\frac{1}{p^+}\right]_+^{(k^+)}$; C3: the last line. In the variables of AABKL: Sec. 11.

9.3.2 Second part

$$\begin{aligned}
\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^4}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,x',y',z,x} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}}{(y'-z)^2\,(x'-y')^2}\,\frac{1}{[p^+\,(x-z)^2 + k^+\,(x-w)^2]}\\
&\left[\frac{\delta_{k'm}\,d_\perp\,p^+}{2\,(z-w)^2\,(p^+ + k^+)^2}\,[(x-z)^2-(x-w)^2] + \frac{p^+\,\delta_{k'm}}{k^+\,(p^+ + k^+)}\left[\frac{(x-z)\cdot(z-w)}{(z-w)^2} - \frac{(x-z)\cdot(x-w)}{2\,(x-w)^2}\right.\right.\\
&\left.\left. - \frac{[(x-z)^2-(x-w)^2]}{2\,(z-w)^2}\right] + \frac{\delta_{k'm}}{(p^+ + k^+)}\left[\frac{(x-w)\cdot(z-w)}{(z-w)^2} + \frac{(x-z)\cdot(x-w)}{2\,(x-z)^2} - \frac{[(x-z)^2-(x-w)^2]}{2\,(z-w)^2}\right]\right.\\
&- \left[\frac{1}{p^+}\right]_+^{(k^+)}\left[\frac{(x-w)^{k'}\,(z-w)^m}{(z-w)^2} + \frac{(x-z)^m\,(x-w)^{k'}}{2\,(x-z)^2}\right] - \frac{p^+}{k^{+\,2}}\left[\frac{(x-z)^{k'}\,(z-w)^m}{(z-w)^2} - \frac{(x-z)^{k'}\,(x-w)^m}{2\,(x-w)^2}\right]\\
&\left. - \frac{1}{k^+}\left[\frac{(x-z)^m\,(z-w)^{k'}}{(z-w)^2} - \frac{(x-z)^m\,(x-w)^{k'}}{2\,(x-w)^2} + \frac{(x-w)^m\,(z-w)^{k'}}{(z-w)^2} + \frac{(x-z)^{k'}\,(x-w)^m}{2\,(x-z)^2}\right]\right]\times\\
&\left[f^{c'd'a}\,U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right]\\
&- \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\left[1-\frac{k^+}{\vee}\right]\int_{w,x',y',z,x} e^{-ik\cdot(y'-w)}\left[\frac{(x-w)\cdot(x'-y')\,(z-w)\cdot(y'-z)}{(z-w)^2\,(x-w)^2\,(y'-z)^2\,(x'-y')^2}\right.\\
&\left.+\frac{(x-z)\cdot(y'-z)\,(x-w)\cdot(x'-y')}{2\,(x-z)^2\,(x-w)^2\,(y'-z)^2\,(x'-y')^2}\right]\left[f^{c'd'a}\,U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\\
&\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right]\\
&+ \frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^4}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\,\frac{1}{k^+\,(p^+ + k^+)}\int_{w,x',y',z,x} e^{-ik\cdot(y'-w)}\,\frac{(x'-z)\cdot(x-z)}{(x'-z)^2\,(x-z)^2}\\
&\left[\frac{(y'-z)\cdot(z-w)}{(y'-z)^2\,(z-w)^2} - \frac{(y'-z)\cdot(x-w)}{2\,(y'-z)^2\,(x-w)^2}\right]\left[f^{c'd'a}\,U^{e'c'}(z)\,U^{d'b}(z)\,\rho^{e'}(x') - f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\\
&\times\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right]
\end{aligned} \tag{9.4}$$

C2: $\left[\frac{1}{p^+}\right]_+^{(k^+)}$; C3: the last line. C5: the measure. In the variables of AABKL: Sec. 11.

9.4 Row IV

NA

9.5 Row V

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'ca}}{4\pi^4}\frac{1}{k^+}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}$$
$$\left[\frac{\delta_{k'm}\,\delta_{ij}}{(p^++k^+)} - \left[\frac{1}{p^+}\right]_+^{(k^+)}\delta_{jm}\,\delta_{ik'} - \frac{1}{k^+}\,\delta_{im}\,\delta_{jk'}\right]\Big[U^{e'c'}(y')\,\rho^{e'}(x') - U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times$$
$$\big[U^{ce}(x)\,U^{ad}(y)\,\rho^e(x)\rho^d(y) - U^{ad}(w)\,U^{ce}(x)\,\rho^e(x)\rho^d(y)\big]$$
$$-\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'ca}}{8\pi^5}\frac{1}{k^+}\log\left[1-\frac{k^+}{\vee}\right]\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}$$
$$\Big[U^{e'c'}(y')\,\rho^{e'}(x') - U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\big[U^{ce}(x)\,U^{ad}(y)\,\rho^e(x)\rho^d(y) - U^{ad}(w)\,U^{ce}(x)\,\rho^e(x)\rho^d(y)\big]$$
$$+\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'ca}}{4\pi^4}\frac{1}{k^+}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\frac{1}{p^++k^+}\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(y-w)\,(x'-z)\cdot(x-z)}{(y'-z)^2\,(y-w)^2\,(x'-z)^2\,(x-z)^2}$$
$$\Big[U^{e'c'}(z)\,\rho^{e'}(x') - U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\big[U^{ce}(x)\,U^{ad}(y)\,\rho^e(x)\rho^d(y) - U^{ad}(w)\,U^{ce}(x)\,\rho^e(x)\rho^d(y)\big] \tag{9.5}$$

C2: $\left[\frac{1}{p^+}\right]_+^{(k^+)}$; C3: the last line. In the variables of AABKL: Sec. 11.

9.6 Row VI

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'd'a}}{4\pi^4}\frac{1}{k^+}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}$$
$$\left[\frac{\delta_{k'm}\,\delta_{ij}}{(p^++k^+)} - \left[\frac{1}{p^+}\right]_+^{(k^+)}\delta_{jm}\,\delta_{ik'} - \frac{1}{k^+}\,\delta_{im}\,\delta_{jk'}\right]\Big[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times$$
$$\big[U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^d(y)\,\rho^e(x) - U^{ad}(y)\,\rho^d(y)\,\rho^b(x)\big]$$
$$-\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'd'a}}{8\pi^5}\frac{1}{k^+}\log\left[1-\frac{k^+}{\vee}\right]\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}$$
$$\Big[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\big[U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^d(y)\,\rho^e(x) - U^{ad}(y)\,\rho^d(y)\,\rho^b(x)\big]$$
$$+\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'd'a}}{4\pi^4}\frac{1}{k^+}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\frac{1}{p^++k^+}\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(y-w)\,(x'-z)\cdot(x-z)}{(y'-z)^2\,(y-w)^2\,(x'-z)^2\,(x-z)^2}$$
$$\Big[U^{e'c'}(z)\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\big[U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^d(y)\,\rho^e(x) - U^{ad}(y)\,\rho^d(y)\,\rho^b(x)\big] \tag{9.6}$$

C2: $\left[\frac{1}{p^+}\right]_+^{(k^+)}$; C3: the last line. In the variables of AABKL: Sec. 11.

10 Result

$$\begin{aligned}
&\frac{d^3N_h}{d^3k} = \int_{k^+/\vee}^1\frac{d\zeta}{\zeta^2}\,D_{h/g}(\zeta,\mu_F)\,\frac{d^3N_g}{d^3k}\Big(\frac{k^+}{\zeta},\frac{\mathbf{k}}{\zeta}\Big),\\
&\frac{d^3N_g}{d^3k} = \mathrm{LO}\big|_{\alpha_s(\mu_R)} + \frac{\alpha_s\beta_0}{4\pi}\log\frac{\mu_R^2}{\mu_F^2}\,\mathrm{LO} + \frac{\alpha_s}{2\pi}N_c\Big[\frac{67}{9}-\frac{\pi^2}{3}\Big]\mathrm{LO} \qquad \text{(running coupling, Sec. 4)}\\
&\qquad + \mathcal{N}\int_{x,x',z,w,w'} e^{-ik(w'-w)}\Big[\log\frac{k^+}{\Lambda}\,\mathcal{J} + \log\frac{\vee}{k^+}\,\mathcal{B}\Big] \qquad \text{(JIMWLK: target; BFKL: projectile)}\\
&\qquad + \frac{d^3N_{\rm DGLAP}}{d^3k} \qquad \text{(Eq. (5.1))}\\
&\qquad + \frac{\alpha_sN_c}{4\pi}\log\Big(1-\frac{k^+}{\vee}\Big)\,\mathrm{LO}\big[\hat{E}_X+\hat{E}_{X'}\big] \qquad \text{(Q1, Eq. (5.2))}\\
&\qquad + \text{finite terms (Sec. 6)} + \text{finite part of } \mathcal{D}\ \text{(Q3)} + \text{four-}\rho\ \text{term (Q2)} + O(\alpha_s^3).
\end{aligned} \tag{10.1}$$

- $\mathcal{J}$, $\mathcal{B}$: the brackets of Secs. 3 and 4 of `Evolution_terms.tex` (JIMWLK⊗LO, BFKL⊗LO). They hold with fixes (A) and (B), `BFKL_final.pdf`. In your split, $\log\frac{\vee}{\Lambda}\mathcal{J}$ (`Evolution_terms.tex`) $+\log\frac{\vee}{k^+}(\mathcal{B}-\mathcal{J})$ (`bfkl.tex`) is the same thing.
- The finite terms contain three-ρ rows. Their fully symmetric part needs $\langle\rho\rho\rho\rangle$, which is zero for a Gaussian projectile.
- The four-ρ finite term (Q2) is not in `Finite_terms_1.tex`.

11 In the variables of AABKL (arXiv:1411.2869)

T. Altinoluk, N. Armesto, G. Beuf, A. Kovner, M. Lublinsky: forward single-inclusive production in the hybrid formalism. The projectile is a collinear parton with PDF $f(x_p)$, LO is $O(\alpha_s^0)$, NLO is one splitting. Their gluon channel is Eq. (4.22): a real term $\int_0^{1-x_p} d\xi$, a virtual term $\int_0^1 d\xi$, and a soft term $\int \frac{d\xi}{\xi}$. Here ξ is the momentum fraction of the unobserved gluon. Their $A^i(y-z) = -\frac{1}{2\pi}\frac{(y-z)^i}{(y-z)^2} \leftrightarrow -\frac{1}{2\pi}\mathcal{K}(y-z)$, and $s_A[y,\bar{y}] \leftrightarrow$ the colour factors in U.

11.1 Group III: the real term

Change of variables. In Rows III (both parts), V, VI put

$$\xi = \frac{p^+}{p^+ + k^+} \in \Big[0,\, 1 - \frac{k^+}{\vee}\Big], \qquad y' = (1-\xi)\,u + \xi\, z, \qquad dp^+\, d^2y' = k^+\, d\xi\, d^2u. \tag{11.1}$$

- The phase becomes $e^{-ik\cdot(y'-w)}e^{-ik\cdot(y'-z)\frac{p^+}{k^+}} = e^{-ik\cdot(u-w)}$. So u is the measured gluon in the conjugate amplitude, z the unobserved gluon, and y' the parent. The parent sits at the momentum-weighted centre $(1-\xi)u+\xi z$ and crosses the target there ($U(y')$). This is AABKL's $s[(1-\xi)\bar{y}+\xi z, \dots]$.
- The kernel $\frac{(y'-z)^m}{(y'-z)^2} = \frac{1}{1-\xi}\frac{(u-z)^m}{(u-z)^2}$ is the splitting kernel in the daughter separation, their $A^i(\bar{y}-z)$.
- The upper limit $\vee - k^+$ becomes $1 - \frac{k^+}{\vee}$: their $1-x_p$ with $x_p \leftrightarrow \frac{k^+}{\vee}$.
- Your energy denominator becomes $p^+(x-z)^2 + k^+(x-w)^2 = \frac{k^+}{1-\xi}D_\xi$, with $D_\xi \equiv \xi(x-z)^2 + (1-\xi)(x-w)^2$.

The plus prescriptions. Define, on $[0, 1-x]$ with $x = \frac{k^+}{\vee}$,

$$\int_0^{1-x} d\xi\, \frac{g(\xi)}{[\xi]_+} \equiv \int_0^{1-x} d\xi\, \frac{g(\xi)-g(0)}{\xi} + g(0)\log(1-x), \qquad \int_0^{1-x} d\xi\, \frac{g(\xi)}{[1-\xi]_\oplus} \equiv \int_0^{1-x} d\xi\, \frac{g(\xi)-g(1)}{1-\xi}. \tag{11.2}$$

$[\xi]_+$ is the usual plus distribution on $[0,1]$, the one of AABKL. $[1-\xi]_\oplus$ subtracts the value at $\xi = 1$ (the measured gluon soft compared with its parent). Both act at fixed u, x, x', y, z, w, on everything to their right. Then (`checks_xi.json`):

- $\int_0^{\vee - k^+} dp^+ \left[\frac{1}{p^+}\right]_+^{(k^+)} F = \int_0^{1-x} d\xi\, \frac{1}{\xi}\Big(\frac{F}{1-\xi} - F(0)\Big)$ exactly (difference 0). Adding your $\log(1-\frac{k^+}{\vee})$ line, which is $-F(0)\log(1-\frac{k^+}{\vee})$ times the same term, gives $\int_0^{1-x}\frac{d\xi}{[\xi]_+}\frac{F}{1-\xi}$. So your $\left[\frac{1}{p^+}\right]_+^{(k^+)}$ (C2) plus your $\log(1-\frac{k^+}{\vee})$ lines are AABKL's $\frac{1}{[\xi]_+}$.
- The $-\frac{1}{k^+}\delta_{im}\delta_{jk'}$ (resp. $-\frac{p^+}{k^{+2}}[\dots]$) term together with the C3 line is $-\int_0^{1-x}\frac{d\xi}{[1-\xi]_\oplus}(\dots)$ (difference 1×10^{-16}). The value at $\xi = 1$ is the integrand of the C3 line (`checks_xi.json`: relative difference $3\times10^{-2}, 3\times10^{-4}, 3\times10^{-6}$ at $1-\xi = 10^{-4}, 10^{-6}, 10^{-8}$). What $[1-\xi]_\oplus$ subtracts, $g(1)\int_0^{1-x}\frac{d\xi}{1-\xi} = g(1)\log\frac{\vee}{k^+}$, is the BFKL term (B).
- Pointwise, $k^+\times$(your p^+ integrand) = (ξ integrand below) for every term of the rows (largest relative difference 1×10^{-14} for Rows V, III(1), VI and 2×10^{-14} for Row III(2), 200 random points).

Rows V, VI, III part 1 (each row = your row with C2 + your $\log(1-\frac{k^+}{\vee})$ line + the C3 line):

$$\begin{aligned}
\frac{d^3N_{\rm V}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'ca}}{8\pi^5}\,\frac{1}{k^+}\int_0^{1-\frac{k^+}{\vee}} d\xi \int_{w,u,z,y,x,x'} e^{-ik\cdot(u-w)}\,\frac{(u-z)^m\,(x'-y')^{k'}\,(y-w)^i\,(x-z)^j}{(u-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\\
&\times\left[\delta_{k'm}\,\delta_{ij} - \frac{1}{[\xi]_+}\,\delta_{jm}\,\delta_{ik'} - \frac{1}{[1-\xi]_\oplus}\,\delta_{im}\,\delta_{jk'}\right]\Big[U^{e'c'}(y')\,\rho^{e'}(x') - U^{c'e'}(x')\,\rho^{e'}(x')\Big]\\
&\times\big[U^{ce}(x)\,U^{ad}(y)\,\rho^e(x)\rho^d(y) - U^{ad}(w)\,U^{ce}(x)\,\rho^e(x)\rho^d(y)\big]\,, \qquad y' = (1-\xi)u+\xi z,
\end{aligned}\tag{11.3}$$

$$\begin{aligned}
\frac{d^3N_{\rm VI}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{8\pi^5}\,\frac{1}{k^+}\int_0^{1-\frac{k^+}{\vee}} d\xi \int_{w,u,z,y,x,x'} e^{-ik\cdot(u-w)}\,\frac{(u-z)^m\,(x'-y')^{k'}\,(y-w)^i\,(x-z)^j}{(u-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\\
&\times\left[\delta_{k'm}\,\delta_{ij} - \frac{1}{[\xi]_+}\,\delta_{jm}\,\delta_{ik'} - \frac{1}{[1-\xi]_\oplus}\,\delta_{im}\,\delta_{jk'}\right]\Big[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\\
&\times\big[U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^d(y)\,\rho^e(x) - U^{ad}(y)\,\rho^d(y)\,\rho^b(x)\big]\,,
\end{aligned}\tag{11.4}$$

$$\begin{aligned}
\frac{d^3N_{\rm III(1)}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{16\pi^5}\,\frac{1}{k^+}\int_0^{1-\frac{k^+}{\vee}} d\xi \int_{w,x',u,z,y,x} e^{-ik\cdot(u-w)}\,\frac{(u-z)^m\,(x'-y')^{k'}\,(y-w)^i\,(x-z)^j}{(u-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\\
&\times\left[\delta_{k'm}\,\delta_{ij} - \frac{1}{[\xi]_+}\,\delta_{jm}\,\delta_{ik'} - \frac{1}{[1-\xi]_\oplus}\,\delta_{im}\,\delta_{jk'}\right]\Big[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\\
&\times\big[U^{ac}(w)\,\{\rho^c(y),\rho^b(x)\} - U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\{\rho^d(y),\rho^e(x)\}\big]\,.
\end{aligned}\tag{11.5}$$

The bracket $\big[\delta_{k'm}\delta_{ij} - \frac{1}{\xi}\delta_{jm}\delta_{ik'} - \frac{1}{1-\xi}\delta_{im}\delta_{jk'}\big]$ is the light-cone three-gluon vertex: its square, summed over polarizations, is $2\big[\frac{1-\xi}{\xi} + \frac{\xi}{1-\xi} + \xi(1-\xi)\big]$, AABKL's bracket. Here it appears once, as an interference with the emission of the measured gluon by $\rho(y)$ ($\mathcal{K}(y-w)$), because the parent itself is emitted by $\rho(x')$.

Row III part 2:

$$\frac{d^3N_{\mathrm{III}(2)}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_0^{1-\frac{k^+}{\vee}} d\xi \int_{w,x',u,z,x} e^{-ik\cdot(u-w)}\,\frac{(u-z)^m\,(x'-y')^{k'}}{(u-z)^2\,(x'-y')^2}\,\frac{1}{D_\xi}$$
$$\times\Bigg\{\delta_{k'm}\,d_\perp\,\frac{\xi(1-\xi)}{2\,(z-w)^2}\left[(x-z)^2-(x-w)^2\right]+\xi\,\delta_{k'm}\left[\frac{(x-z)\cdot(z-w)}{(z-w)^2}-\frac{(x-z)\cdot(x-w)}{2\,(x-w)^2}-\frac{(x-z)^2-(x-w)^2}{2\,(z-w)^2}\right]$$
$$+(1-\xi)\,\delta_{k'm}\left[\frac{(x-w)\cdot(z-w)}{(z-w)^2}+\frac{(x-z)\cdot(x-w)}{2\,(x-z)^2}-\frac{(x-z)^2-(x-w)^2}{2\,(z-w)^2}\right]$$
$$-\frac{1-\xi}{[\xi]_+}\left[\frac{(x-w)^{k'}\,(z-w)^m}{(z-w)^2}+\frac{(x-z)^m\,(x-w)^{k'}}{2\,(x-z)^2}\right]-\frac{\xi}{[1-\xi]_\oplus}\left[\frac{(x-z)^{k'}\,(z-w)^m}{(z-w)^2}-\frac{(x-z)^{k'}\,(x-w)^m}{2\,(x-w)^2}\right]$$
$$-\left[\frac{(x-z)^m\,(z-w)^{k'}}{(z-w)^2}-\frac{(x-z)^m\,(x-w)^{k'}}{2\,(x-w)^2}+\frac{(x-w)^m\,(z-w)^{k'}}{(z-w)^2}+\frac{(x-z)^{k'}\,(x-w)^m}{2\,(x-z)^2}\right]\Bigg\}$$
$$\times\left[f^{c'd'a}\,U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right], \tag{11.6}$$

with $D_\xi=\xi(x-z)^2+(1-\xi)(x-w)^2$; the plus prescriptions act also on $\frac{1}{D_\xi}$, $(x'-y')$ and $U(y')$. Your p^+ coefficients become polynomials in ξ: $\frac{p^+}{(p^++k^+)^2}\to\frac{\xi(1-\xi)}{k^+}$, $\frac{p^+}{k^+(p^++k^+)}\to\frac{\xi}{k^+}$, $\frac{1}{p^++k^+}\to\frac{1-\xi}{k^+}$, $\frac{p^+}{k^{+2}}\to\frac{\xi}{k^+(1-\xi)}$. At $\xi=0$, $\frac{1-\xi}{D_\xi}=\frac{1}{(x-w)^2}$, the weight of your $\log(1-\frac{k^+}{\vee})$ line. At $\xi=1$, $\frac{\xi}{D_\xi}=\frac{1}{(x-z)^2}$, the weight of the C3 line.

11.2 Group I Row V: the virtual term

With $\xi=\frac{p^+}{k^+}\in[0,1]$, your $\left[\frac{k^+}{p^+}\right]_+$ and $\left[\frac{1}{q^+}\right]_+$ on $[0,k^+]$ are $\frac{1}{[\xi]_+}$ on $[0,1]$, AABKL's virtual $\int_0^1 d\xi$. The first part, for $\Lambda\to 0$:

$$\frac{d^3N_1}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{ig^4}{32\pi^5}\,\frac{1}{k^+}\int_0^1 d\xi\int_{x,z,y,x',y',w'} e^{-ik\cdot(w'-[(1-\xi)z+\xi x])}\,\frac{(x'-w')^i\,(z-x)^m\,(y-z)^k\,(y'-x)^j}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2}\left[\delta_{im}\,\delta_{jk}-\frac{1}{[\xi]_+}\,\delta_{jm}\,\delta_{ik}\right]$$
$$\times\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\left[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^e(y),\rho^b(y')\}-f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^d(y),\rho^e(y')\}\right]$$
$$-\frac{1}{(2\pi)^3}\,\frac{ig^4}{32\pi^5}\,\frac{1}{k^+}\int_0^1\frac{d\xi}{[\xi]_+}\int_{x,z,y,x',y',w'} e^{-ik\cdot(w'-[(1-\xi)x+\xi z])}\,\frac{(x'-w')\cdot(y'-x)\,(z-x)\cdot(y-z)}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2}$$
$$\times\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\left[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^e(y),\rho^b(y')\}-f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^d(y),\rho^e(y')\}\right]. \tag{11.7}$$

The measured gluon of the amplitude sits at the weighted centre of the two gluons of the loop, as in their $s[y+\xi(\bar y-z),\dots]$. In 2a and 2b, the coefficients $\frac{p^+(k^+-p^+)}{k^{+2}}$, $\frac{k^+-p^+}{k^+}$, $\frac{p^+}{k^+}$, $\left[\frac{k^+}{p^+}\right]_+$ are $\xi(1-\xi)$, $1-\xi$, ξ, $\frac{1}{[\xi]_+}$, and the energy denominator is $k^+[\xi(y-x)^2+(1-\xi)(y-z)^2]$.

11.3 Your result in the layout of AABKL Eq. (4.22)

$$\frac{d^3N_g}{d^3k} = \mathrm{LO} + \text{running coupling (Sec. 4)}$$
$$+\underbrace{\int_0^{1-\frac{k^+}{\vee}} d\xi\,\Big[\text{Group III Rows III, V, VI (above)}\Big]}_{\text{AABKL: }\int_0^{1-x_p}d\xi\,\frac{x_p}{1-\xi}f(\frac{x_p}{1-\xi})\,[\dots]\,\left(-\frac{N_c^3}{N_c^2-1}A\cdot A\{sss+\dots-\frac{1}{N_c^2}X\}\right)}+\underbrace{\frac{d^3N_{\mathrm{DGLAP}}}{d^3k}}_{\text{AABKL: }2C_A[A\cdot A-(AA)_{\mu^2}](1-\xi)^2 s_A[(1-\xi)y,(1-\xi)\bar y]}$$
$$+\underbrace{\Big[\text{Group II Row I part 2, Row IV line 3; Group III Rows I, II}\Big]}_{\text{AABKL: }2C_A[A\cdot A-(AA)_{\mu^2}]s_A[y,\bar y]\text{ (before the target); no counterpart}}$$
$$+\underbrace{\int_0^1 d\xi\,\Big[\text{Group I Row V}\Big]+\Big[\text{Group I Rows II, III, IV}\Big]}_{\text{AABKL: }-x_p f\int_0^1 d\xi\,[\dots]\,\left(C_A[AA+AA-2(AA)_{\mu^2}]s_A+\frac{N_c^3}{2(N_c^2-1)}AA\{\dots\}\right)}$$
$$+\underbrace{\mathcal{N}\int\log\frac{k^+}{\Lambda}\,\mathcal{J}}_{\text{AABKL: }\int\frac{d\xi}{\xi}[A_\xi-A_\xi]^2\{\cdots-2C_A s_A\}}+\underbrace{\mathcal{N}\int\log\frac{\vee}{k^+}\,\mathcal{B}}_{\text{AABKL: none}}+\mathrm{Q1}+\mathrm{Q2}+\mathrm{Q3}. \tag{11.8}$$

AABKL, Eq. (4.22)	yours	comparison
real, $\int_0^{1-x_p} d\xi$; parent at $(1-\xi)\bar{y}+\xi z$	Group III Rows III, V, VI; $\int_0^{1-k^+/\vee} d\xi$, parent $y'=(1-\xi)u+\xi z$	same variables, same limit, same parent position. Plus prescription at $\xi\to 0$: theirs $\frac{1}{[\xi]_+}$, yours $\frac{1}{[\xi]_+}$ (C2 + your $\log(1-\frac{k^+}{\vee})$ lines). At $\xi\to 1$: theirs is cut off by $f(\frac{x_p}{1-\xi})\to f(1)=0$, yours by $[1-\xi]_\oplus$ (C3), whose subtraction is BFKL. Kernels: your parent is emitted by $\rho(x')$, so the vertex appears once, in interference with emission by $\rho(y)$
after-target collinear term, $(1-\xi)^2 s_A[(1-\xi)y,(1-\xi)\bar{y}]$	DGLAP, Group II Row IV lines 1, 2 (C1)	the same: splitting function ($\xi\to 1-\xi$), plus prescription, $\log(r^2\mu^2)$, scheme constant $R=c_0$ (`checks_rc.json`)
before-target collinear term, $s_A[y,\bar{y}]$	Group II Row I part 2 (colour $ff\,U(w')U(w)$)	same topology and colour; theirs collinear divergent (PDF), yours finite (the parent is emitted at $\lvert x-w\rvert$)
virtual, $-x_p f\int_0^1 d\xi$	Group I Row V (1, 2a, 2b finite)	same $\int_0^1 d\xi\frac{1}{[\xi]_+}$, weighted positions $(1-\xi)z+\xi x$; colour $-ff\,U(x)U(z)+N_cU(x)$ as their C_A and $\frac{N_c^3}{N_c^2-1}$ terms
virtual UV, $\int_0^1 d\xi[\dots]=-\frac{11}{6}$	Group I Row II part 1 ($\frac{11}{3}$), Row V 2b UV ($-\frac{11}{6}$)	theirs: $2C_A\cdot\frac{11}{6}=\beta_0$ enters P_{gg}; yours: the two cancel and $\frac{\beta_0}{2}\delta(1-\xi)$ comes from the coupling of LO
soft term, $\int\frac{d\xi}{\xi}[A_\xi-A_\xi]^2$	JIMWLK, $\log\frac{k^+}{\Lambda}\mathcal{J}$	same real−virtual (BK/JIMWLK) structure for the soft unobserved gluon; theirs cut by the Ioffe time, yours by Λ
—	BFKL, $\log\frac{\vee}{k^+}\mathcal{B}$	none: their parent distribution vanishes at $\xi\to 1-x_p$
—	running coupling: RG term, $\frac{67}{9}-\frac{\pi^2}{3}$	none: their LO is α_s^0
—	Group II Row IV line 3	linear polarization; none (unpolarized incoming gluon)
—	Group I Rows III, IV, Group II Row I part 1, Group III Rows I, II	emission by the charges around the splitting; none (no colour charges in their projectile)
—	three- and four-ρ terms, $V_\vee$, Q1	none: one parton; their virtual term has no x_p limit

In short.

- DGLAP is the same as AABKL's: the splitting function, the plus prescription, the kinematic limit, the logarithm and the scheme constant.
- Group III and Group I Row V, written in ξ, have AABKL's variables, limits, parent positions and plus prescriptions.
- The kernels differ because your parent gluon is emitted by a colour charge, so there is no collinear divergence to absorb into a PDF.
- The terms with no counterpart are: running coupling, BFKL, linear polarization, emission by the charges around the splitting, and three- and four-ρ terms.

12 Open questions

Q1 The pole $\propto\log(1-\frac{k^+}{\vee})$, Eq. (5.2), is neither DGLAP nor running coupling. It vanishes for $k^+\ll\vee$.

Q2 The four-ρ finite term is not in `Finite_terms_1.tex`.

Q3 The finite part of the $\mathcal{D}$ region (C4), which replaces the red terms of Group I Row III.

13 Files

`input/Finite_terms.tex`	your file
`Finite_terms_corrected.tex/.pdf`	your file with C1–C5 (red/blue) and notes on the DGLAP and running-coupling terms
`compare_file.py` → `.json`	Sec. 2
`edge_check.py` → `.json`	C4: Row III at $z\to w$ and on the circle
`checks_rc.py` → `.json`	running coupling, poles, DGLAP, AABKL constants
`checks_xi.py` → `.json`	C3 and the ξ form (Sec. 11)
`make_corrected.py`, `make_note.py`, `static.tex`	the two documents